TCAN1051HVD

DevLab: TCAN1051HVD CAN Transceiver Module

v1.0

2025-10-02
Rev. A

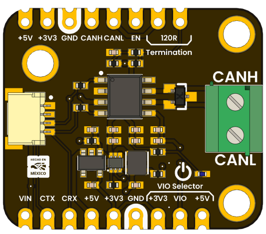
Professional electronic component

PRODUCT OVERVIEW

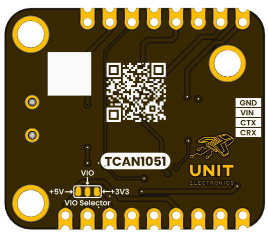
The TCAN1051HVD is a high-speed CAN transceiver designed for automotive and industrial applications. It provides a robust interface between a CAN controller and the physical CAN bus, ensuring reliable data transmission in harsh environments. This module is ideal for use in electric vehicles, industrial automation, and other applications requiring reliable CAN communication. ### Quick Setup PRODUCT WIKI DATASHEET BUY NOW GETTING STARTED | Feature / Component | Detailed Description | | -------------------------------- | -- | | Communication | Integrated high-speed CAN (Controller Area Network) interface, ensuring robust and reliable data transfer in demanding environments. | | Operating Voltage | Supports both 3.3V and 5V logic levels, making it adaptable to a wide range of microcontrollers and external systems. | | Input Port | JST 1.0 mm 4-pin connector providing a compact and secure interface for supplying power and CAN signals. | | IC1 – TCAN1051HVD | Automotive-grade CAN transceiver IC, designed for high noise immunity, ESD protection, and reliable long-distance communication. | | IC2 – AP2112K | 3.3V low dropout (LDO) regulator, delivering a clean and stable voltage supply to sensitive components. | | IC3 – TPS61023 | High-efficiency 5V boost converter, enabling stable power delivery even from lower input voltages. | | L1 – Power LED | On-board LED indicator that provides instant visual feedback of power status. | | JP1 / JP2 | Dual rows of 2.54 mm castellated holes (side 1 and 2), allowing flexible mounting, soldering, or integration into custom boards. | | J1 – CAN Data Port | Dedicated JST 1.0 mm pitch connector optimized for reliable CAN bus data line interfacing. | | J2 – Screw Terminal | Sturdy terminal block that allows secure, tool-based connection to the CAN bus wiring. | | SB1 – Logic Voltage Selector | Configurable solder bridge for selecting the appropriate logic voltage (VIO), ensuring compatibility with the host system. |

PRODUCT VIEWS

TOP VIEW



BOTTOM VIEW



Component placement and connectors

Underside components and connections

KEY TECHNICAL SPECIFICATIONS

🔌 CONNECTIVITY

Primary Interface:	**GPIO (Interrupt)**
Connector Type:	**JST 4-pin 1.0mm**
Logic Levels:	**VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION	NOTES
`Power Supply`	3.3V or 5V

COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	CAN Transceiver
IC2	AP2112K 3.3V Regulator
IC3	TPS61023 5V Regulator
L1	Power On LED
JP1	2.54 mm Castellated Holes
JP2	2.54 mm Castellated Holes
J1	JST 1 mm pitch for CAN Data
J2	Terminal Screw Block for CAN Bus

KEY FEATURES

Touch-only sensing
No physical press required – reacts to proximity of a finger.

Auto-calibration
Compensates for environmental changes and drift.

On-board pull-up/down
Ensures clean digital output.

JST PH-2.0 connector
Quick-disconnect cable interface.

Fast response
< 80 ms touch detection time.

Selectable modes
Momentary or toggle output (via solder-jumper on the board).

Mounting holes
Two M3 screw holes for easy panel integration.

🎯 Key Applications
Automotive systems, Robotics, Home automation, IoT devices

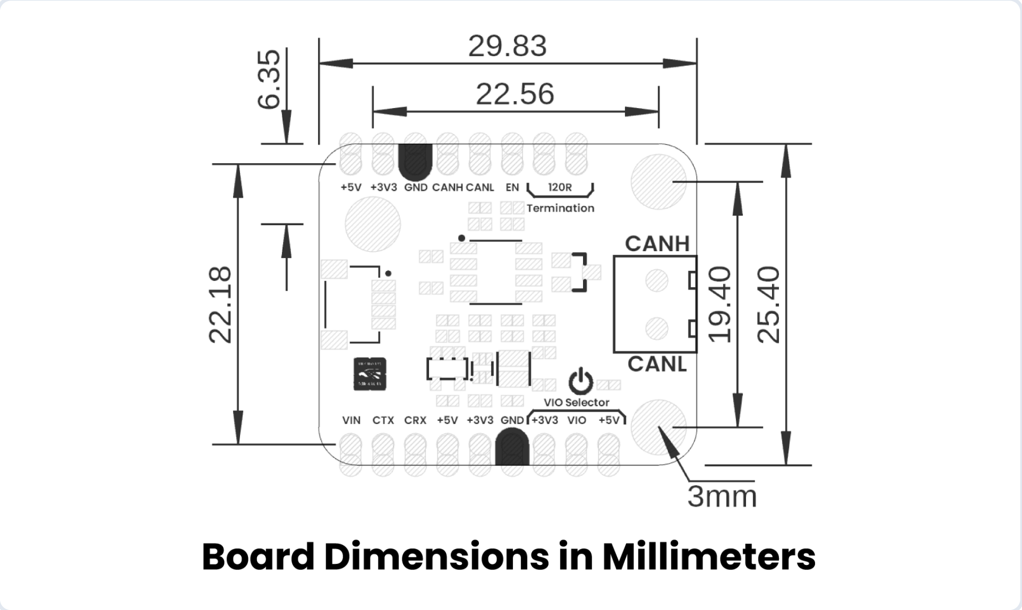
ADDITIONAL TECHNICAL INFORMATION

OVERVIEW

FEATURE / COMPONENT	DETAILED DESCRIPTION
Communication	Integrated **high-speed CAN (Controller Area Network) interface**, ensuring robust and reliable data transfer in demanding environments.
Operating Voltage	Supports both **3.3V and 5V logic levels**, making it adaptable to a wide range of microcontrollers and external systems.
Input Port	**JST 1.0 mm 4-pin connector** providing a compact and secure interface for supplying power and CAN signals.
IC1 – TCAN1051HVD	Automotive-grade **CAN transceiver IC**, designed for high noise immunity, ESD protection, and reliable long-distance communication.
IC2 – AP2112K	**3.3V low dropout (LDO) regulator**, delivering a clean and stable voltage supply to sensitive components.
IC3 – TPS61023	High-efficiency **5V boost converter**, enabling stable power delivery even from lower input voltages.
L1 – Power LED	On-board LED indicator that provides **instant visual feedback of power status**.
JP1 / JP2	Dual rows of **2.54 mm castellated holes** (side 1 and 2), allowing flexible mounting, soldering, or integration into custom boards.
J1 – CAN Data Port	**Dedicated JST 1.0 mm pitch connector** optimized for reliable CAN bus data line interfacing.
J2 – Screw Terminal	Sturdy **terminal block** that allows secure, tool-based connection to the CAN bus wiring.
SB1 – Logic Voltage Selector	**Configurable solder bridge** for selecting the appropriate logic voltage (VIO), ensuring compatibility with the host system.

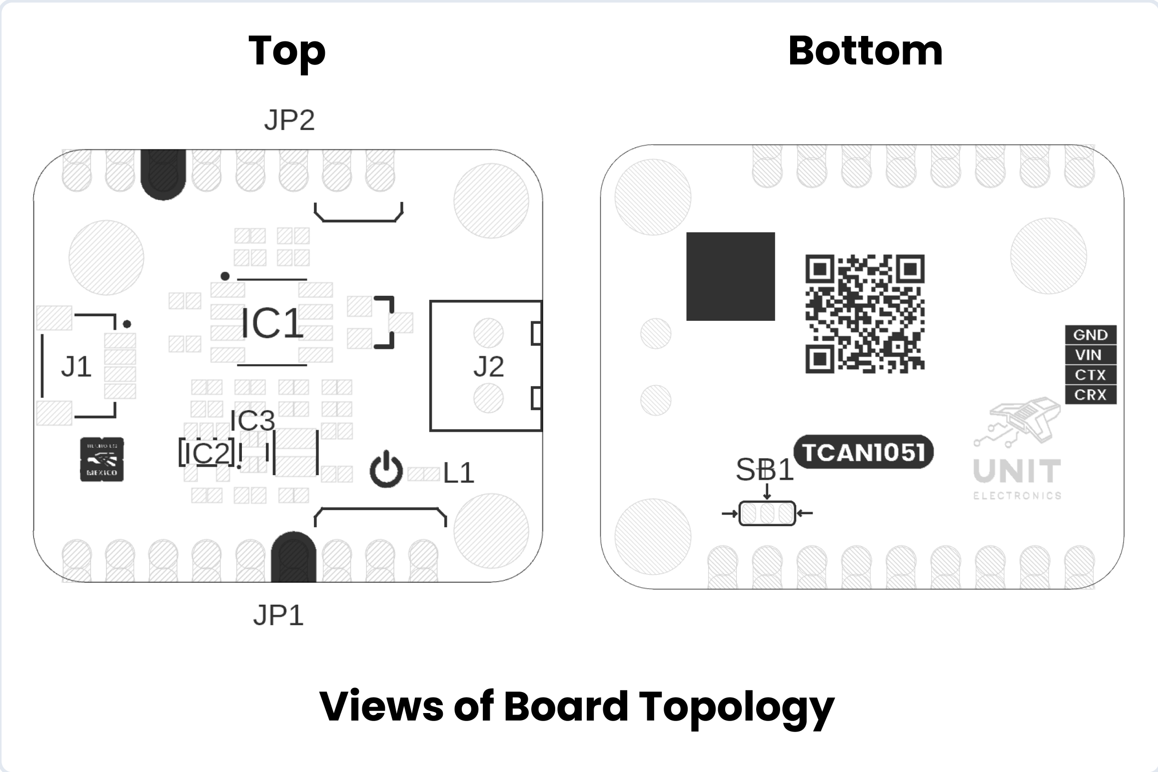
HARDWARE DOCUMENTATION

MECHANICAL DIMENSIONS



Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY



Connection topology and system integration diagram

Click image to open in full size

COMPONENT REFERENCE

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IC1	CAN Transceiver
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J1	JST 1 mm pitch for CAN Data
J2	Terminal Screw Block for CAN Bus
SB1	Solder Bridge for VIO

CIRCUIT SCHEMATIC

Complete circuit schematic showing all component connections

View Complete Schematic PDF

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION	NOTES
Power Supply	3.3V or 5V
Ground	Common ground for all components

PIN CONFIGURATION LAYOUT

Physical connector layout and pin positioning

Pin Configuration Layout

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration

TCAN1051HVD v1.0
Professional Technical Datasheet

Date: 2025-10-02
For commercial distribution